

Duncan Miller

Graduate Student, Department of Biology, Yang Lab, Temple University
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EDUCATION

Master of Science in Molecular and Cellular Biology 2026-Present

Temple University | College of Science and Technology | Department of Biology

Bachelor of Arts in Biochemistry and Molecular Biology 2026

The College of Wooster

Honor: Beta Beta Beta National Biological Honor Society

RESEARCH EXPERIENCE

Graduate Researcher 2026-Present

Cellular Imaging and Super-Resolution Microscopy | Temple University | Yang Lab

Developing a comparative fixation strategy for future two- and three-dimensional PALM mapping of intrinsically disordered FG-nucleoporins. Designing protocols that compare paraformaldehyde, paraformaldehyde-glutaraldehyde, and glyoxal while controlling concentration, pH, temperature, exposure time, washing, and quenching. Reviewing microscopy and fixation literature to identify potential structural artifacts and select conditions for quantitative cellular imaging.

Independent Study Researcher 2025-2026

Microbiology and Molecular Biology | The College of Wooster | Strand Lab

Thesis: *Nutrient-Dependent Phage-Host Dynamics in Biofilm-Forming Pseudomonas*

Investigated how nutrient limitation altered bacteriophage-host dynamics in *Pseudomonas chlororaphis* as a nonpathogenic model for studying *Pseudomonas aeruginosa*. Measured bacterial growth, phage activity, and biofilm formation using microbial culture, plaque assays, growth-curve analysis, and crystal-violet assays. Analyzed relationships among nutrient availability, infection outcomes, and microbial physiology.

PROFESSIONAL EXPERIENCE

Health Coach Intern 2023-2026

Wooster Community Hospital

Prepared and analyzed patient health reports, communicated patient progress to supervising medical staff, and educated patients about evidence-based diet and exercise strategies in clear, accessible language.

Dental Assistant Intern 2022

Santa Rosa Dental Group

Assisted dentists during clinical procedures, prepared instruments, and maintained sterile workspaces.